

NEURAL HARDCORE

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NEURAL HARDCORE
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BRIEF

This project explores how Gabber dance culture can be translated into architectural space through computational design. Starting from a dataset of Gabber dance videos, we use body tracking to analyse dancers' movements, including the trajectories of the feet, hands, head and other body parts, while music intensity is used to describe changes in rhythm and energy. These movement features are classified and transformed into different architectural elements, such as ground surfaces, steps, walls, handrails and roofs. An interactive web platform is developed as part of this system, allowing users to upload or record movements, search for similar dance clips, compare body motions and generate corresponding spatial elements.

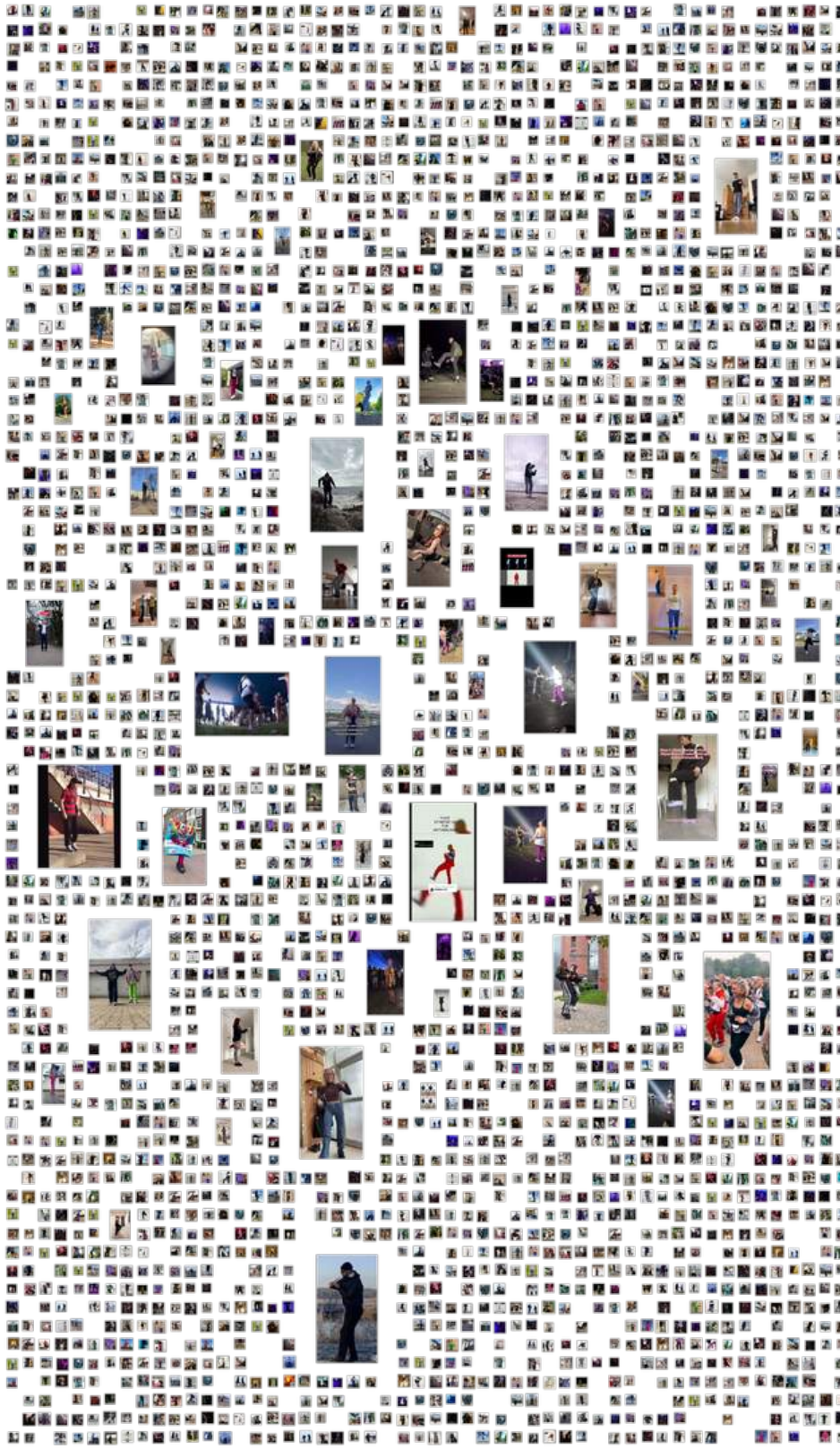
The project is tested at Maassilo in Rotterdam, a site closely connected to the industrial and cultural context of Gabber. Dance movement and music data are mapped onto the site to create spatial heatmaps, movement paths and different levels of activity. These mappings guide the distribution, scale and combination of architectural elements, gradually forming a larger spatial system for circulation, gathering, performance and interaction. Ultimately, these movement-derived elements are integrated to form a Gabber club. Rather than treating Gabber as a temporary event or visual reference, the project uses embodied movement as a design material, exploring how a subculture can leave a spatial trace and become part of an evolving architectural landscape.



https://zihui-gavin.tumblr.com/

DANCE DATASET

The movement dataset was assembled from YouTube and TikTok, focusing on Hakken dance videos featuring different dancers, locations and recording conditions. More than 500 videos were collected and segmented into approximately 9,000 two-second clips. Each clip functions as an individual movement sample for body tracking, movement classification, similarity matching and architectural translation.



DANCE DATASET

A GAN was trained on the collected Gabber album-cover archive. The same 16 latent samples were rendered at selected checkpoints, allowing the model's visual development to be compared directly. Early noise gradually develops into recurring colour fields, graphic structures and distorted motifs associated with the source archive. The loss curves provide a compact record of the changing balance between the generator and discriminator across 200 training epochs.

GABBER DCGAN / TRAINING PROCESS

01 / EARLY — EPOCH 0010



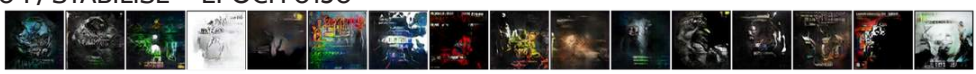
02 / FORMATION — EPOCH 0050



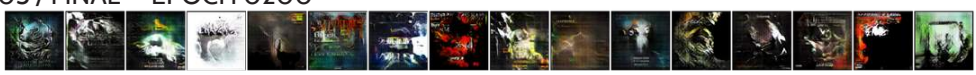
03 / STRUCTURE — EPOCH 0100



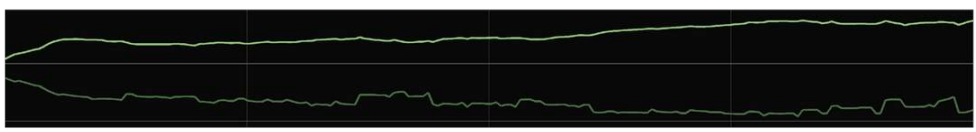
04 / STABILISE — EPOCH 0150



05 / FINAL — EPOCH 0200



GAN LOSS / 15-EPOCH MOVING AVERAGE



Training process video

BODY TRACKING

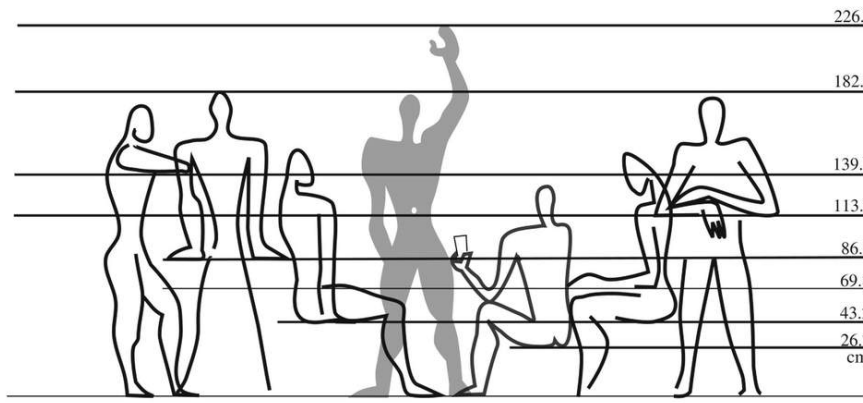
MODULOR / THE BODY AS A 1:1 DESIGN UNIT

HUMAN PROPORTION AS ARCHITECTURAL MEASURE

Le Corbusier's Modulor translates bodily dimensions into an architectural proportioning system. It combines human measurements with a geometric sequence, using reference heights such as the 1.83 m standing figure and the 2.26 m raised-arm reach.

APPLICATION IN THIS PROJECT

The dancers are reconstructed at full 1:1 scale. Joint distances become reach, step, clearance, handrail height and spatial boundary. Generated forms are therefore checked against the size and movement capacity of a real body.

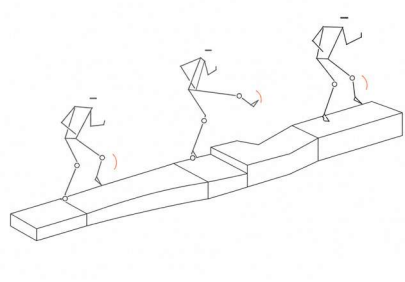


TRACKED BODY / 1:1 -
REACH · STEP · CLEARANCE - AR-
CHITECTURAL DIMENSION

ELEMENT CATALOGUE



foot trajectories transform into ground lines

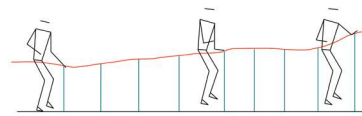


leg movements generate steps



wrist trajectories are lofted into a surface

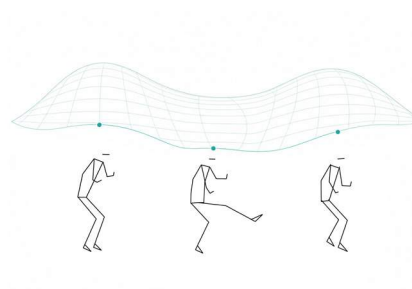
The body is divided into key regions—the feet, hands, shoulders and head—to investigate how their movement trajectories can be translated into architectural elements. These trajectories are subsequently transformed into a spatial vocabulary of ground paths, steps, surfaces, handrails, walls and roofs.



wrist movement generates the handrail



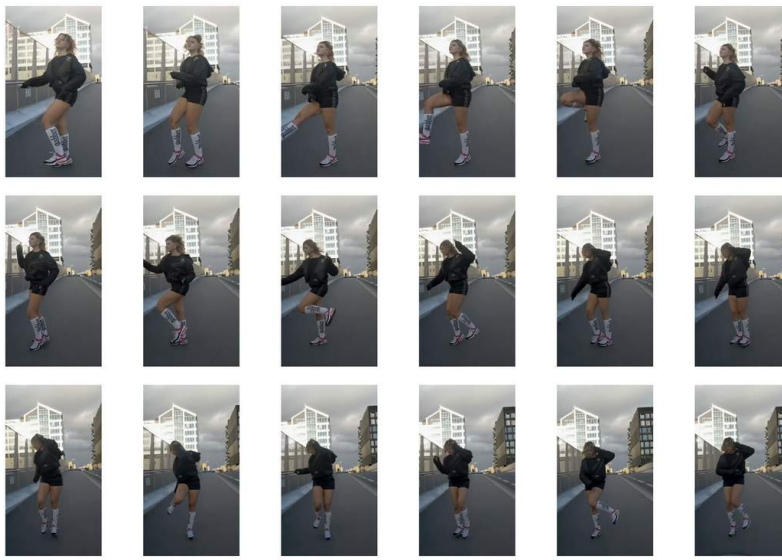
touching wrist trajectories define the wall boundary

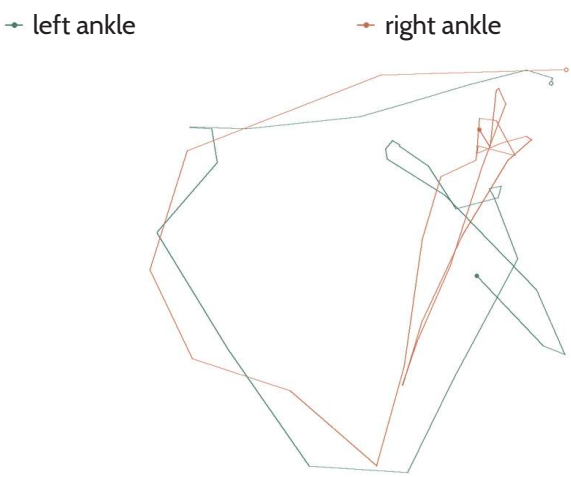


head movement shapes the roof canopy

ELEMENT GENERATION

Foot-Path Tube Generation





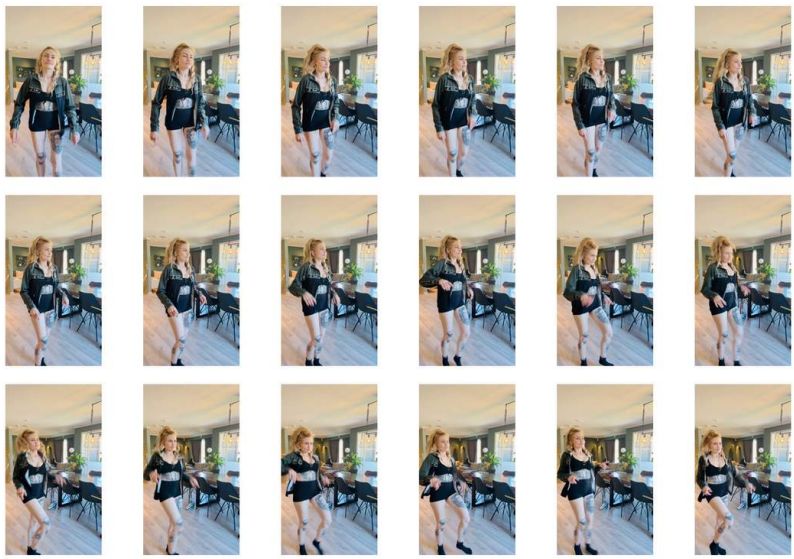
video and generation process link



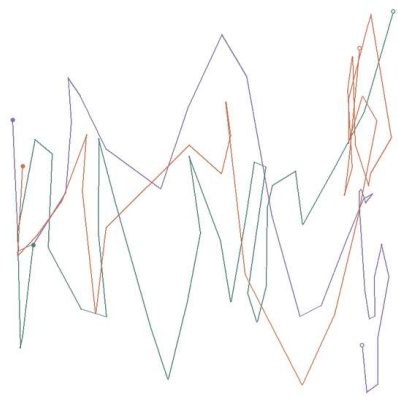
The trajectories of the dancer's left and right feet grow simultaneously and are transformed into two continuous spatial tubes.

ELEMENT GENERATION

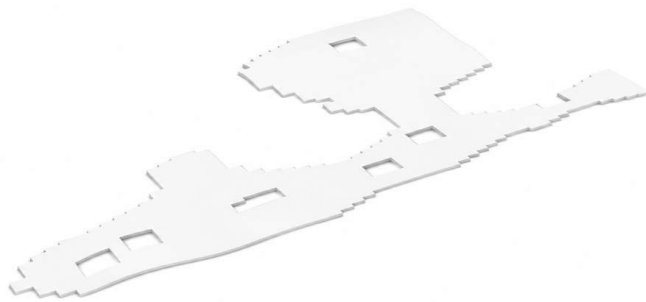
Head-Field Roof Generation



head
left shoulder
right shoulder

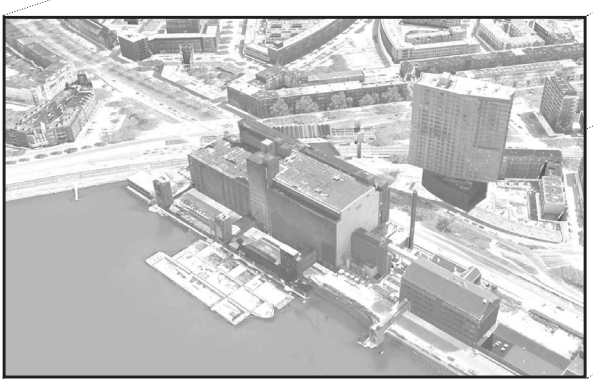


video and generation process link

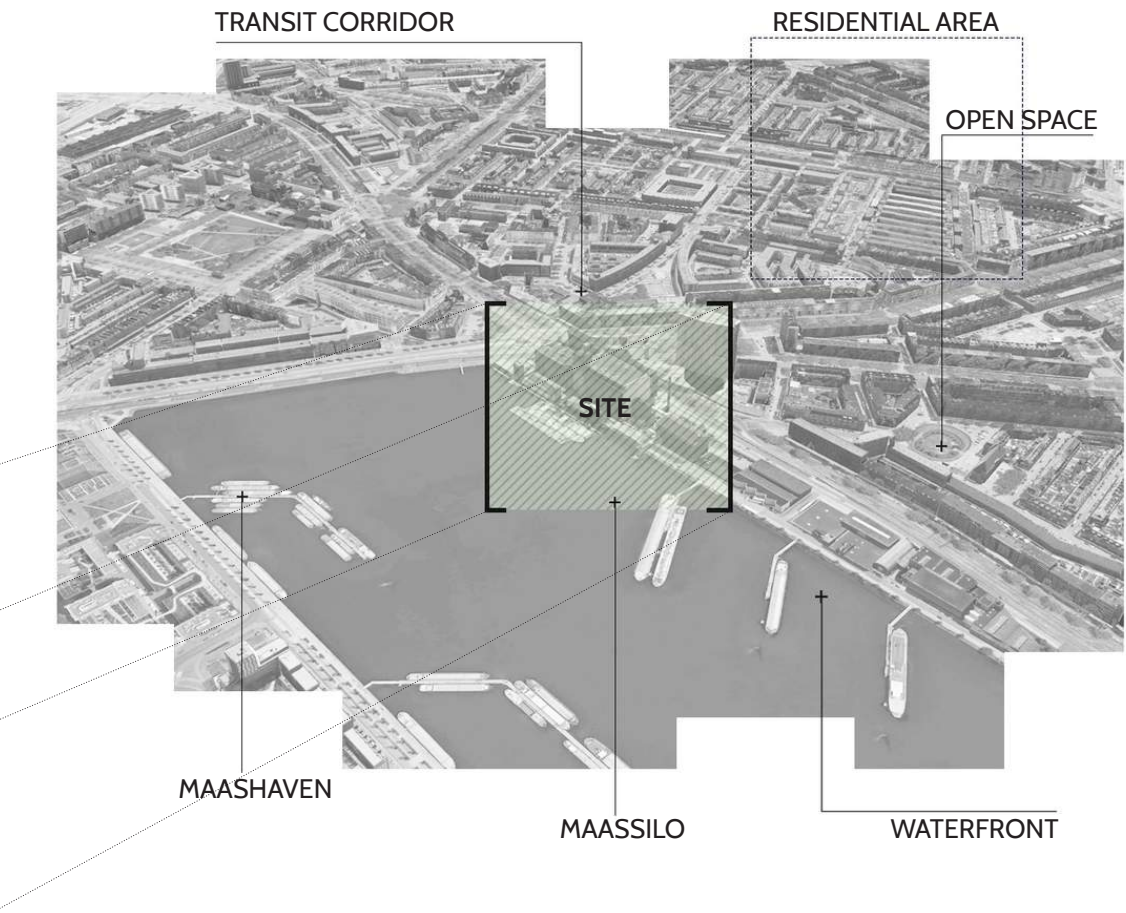


Head position determines where the roof rises, head height controls the degree of deformation, and shoulder width defines the area of influence. These local deformations are blended into a continuous canopy, while peaks in music intensity generate openings across its surface.

SITE CONTEXT

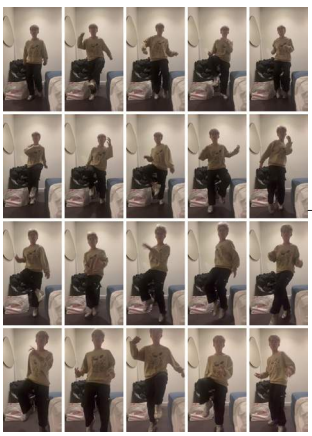


MAASSILO


TRANSIT CORRIDOR
RESIDENTIAL AREA
OPEN SPACE
SITE
MAASHAVEN
MAASSILO
WATERFRONT

DANCE MATCH WEB

IMPORT VIDEO



video link

DTW MOTION MATCHING

Full Body

Foot-Path Tube

Landing-Based Step

Dual-Wrist Lofted Surface

Wrist-Elbow Handrail

Shoulder-Elbow Boundary

Head-Field Roof

recorded video

DTW match

DANCE IN DATASET















video link

MODELS

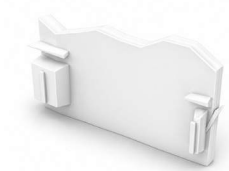
Architectural elements are selected by calculating routing weights for each action clip and applying predefined thresholds.







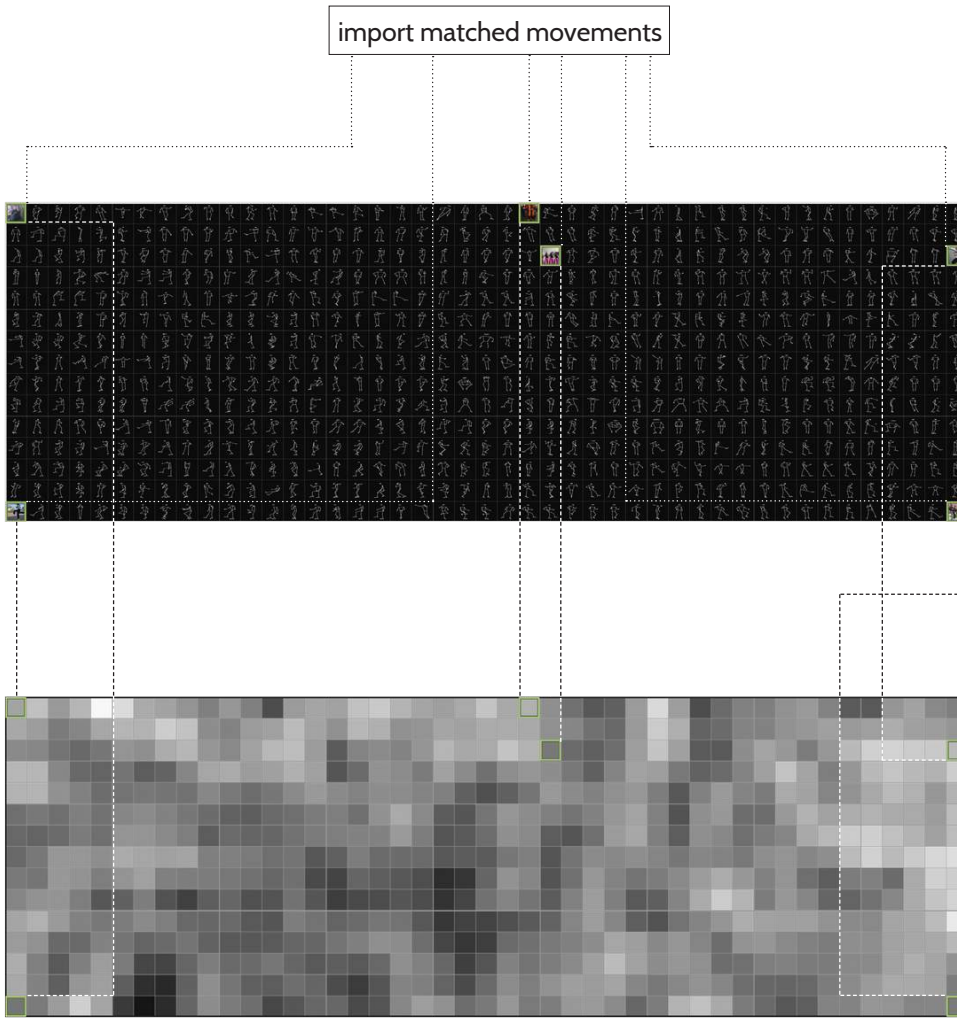






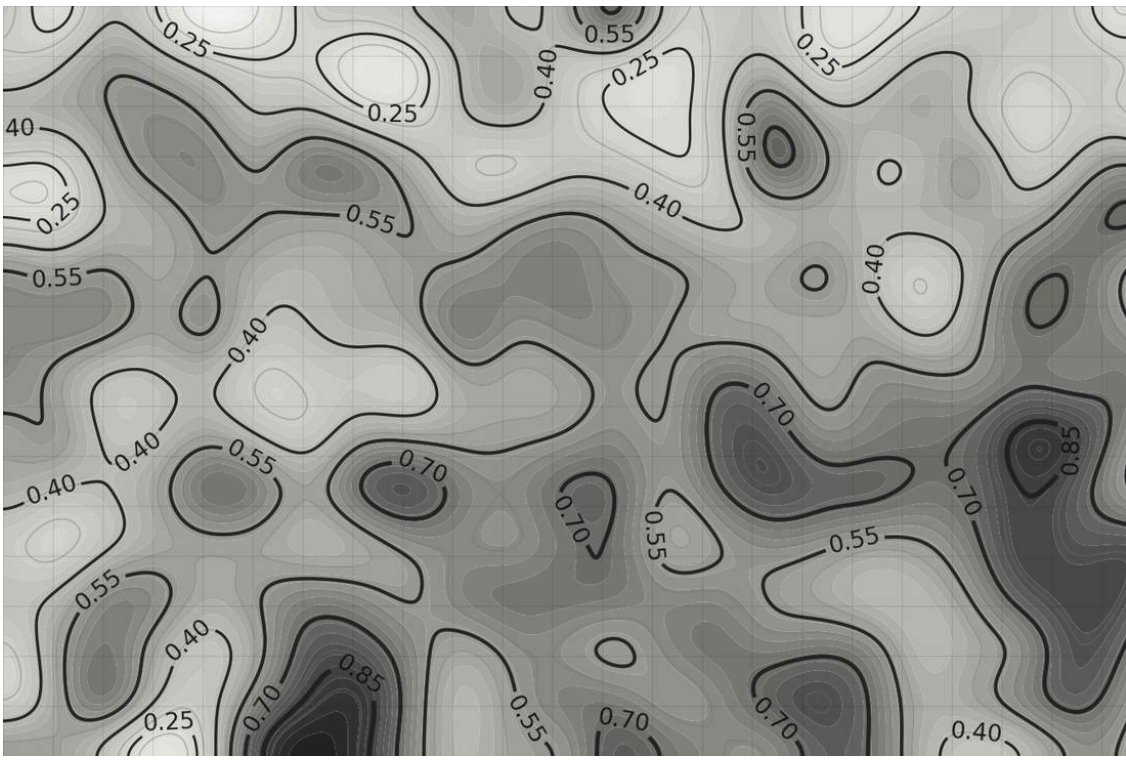
MOVEMENT HEATMAP

Recorded dance movements are matched with similar video clips using DTW and imported into the larger site map. The matched clips are distributed across the map to create a movement-based spatial field. Six MediaPipe foot landmarks—the left and right ankles, heels and foot-index points near the toes—are then tracked and accumulated over time to generate a movement heatmap, revealing different levels of foot-activity density across the site.

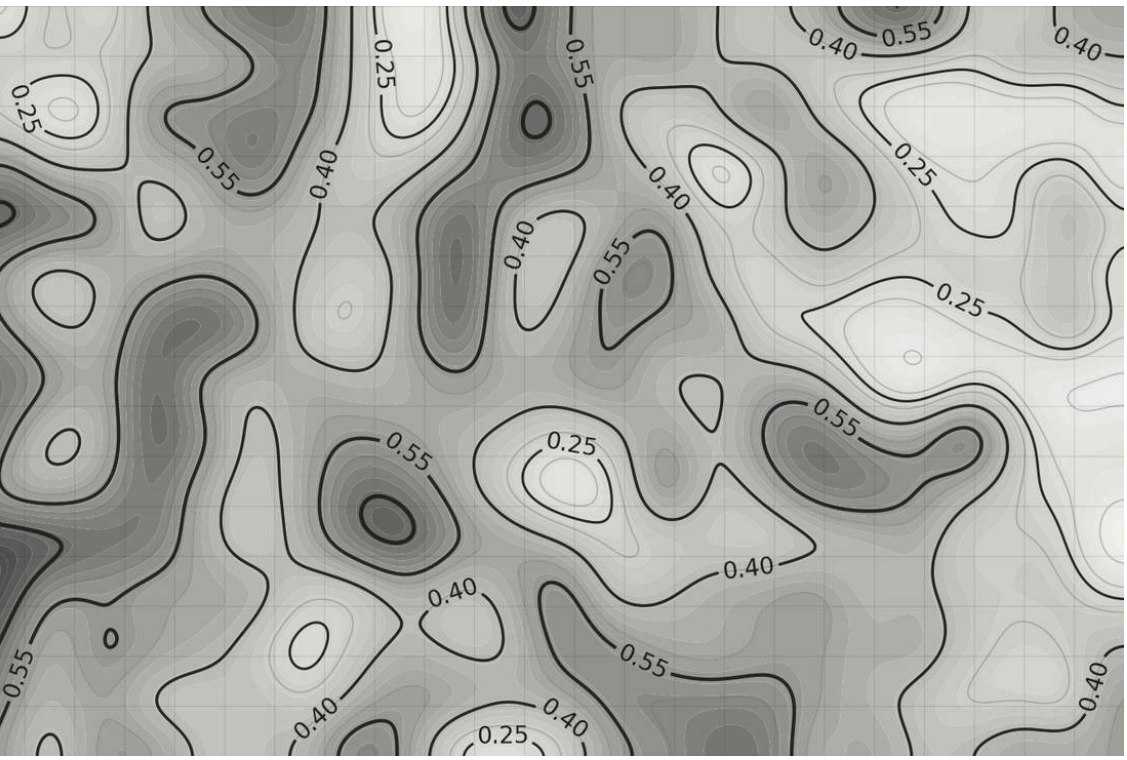


heatmap web display link

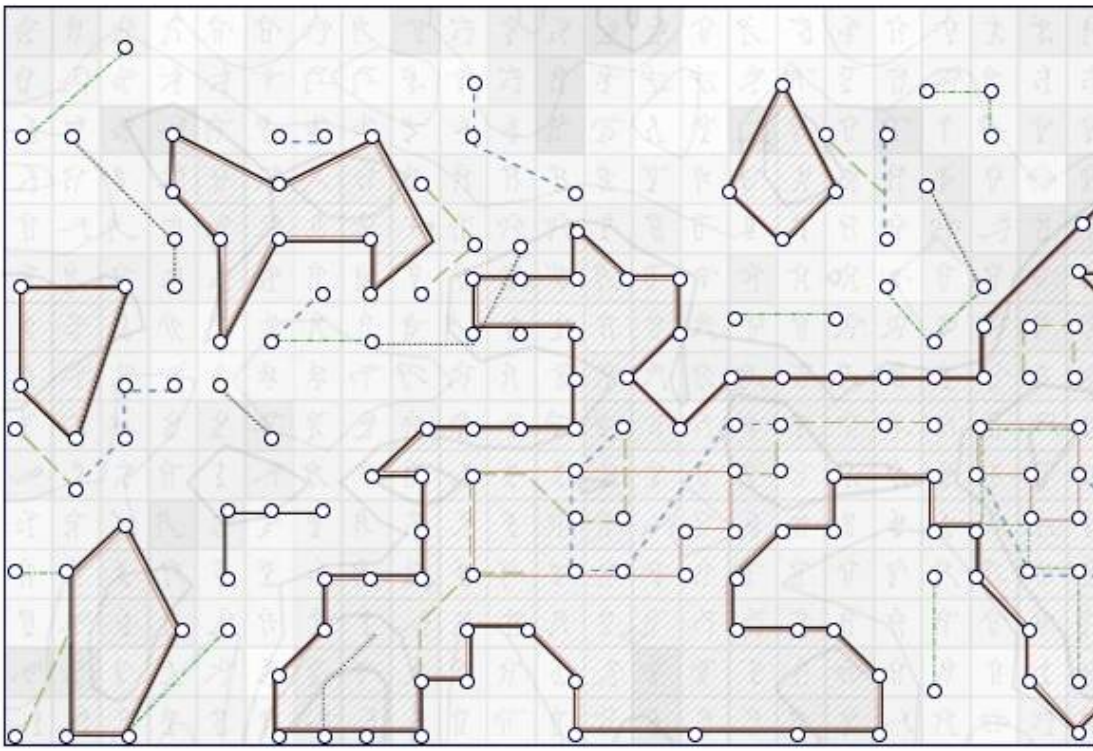
SPATIAL INTENSITY



The Foot Movement Heatmap is translated into functional zones using movement-intensity thresholds. Areas with values of 0.70 or above become stages. Values from 0.55 to below 0.70 become buildings, while values from 0.40 to below 0.55 become platforms. Areas below 0.40 remain as green landscape. In this way, the position and boundary of each function are determined directly by the movement data.

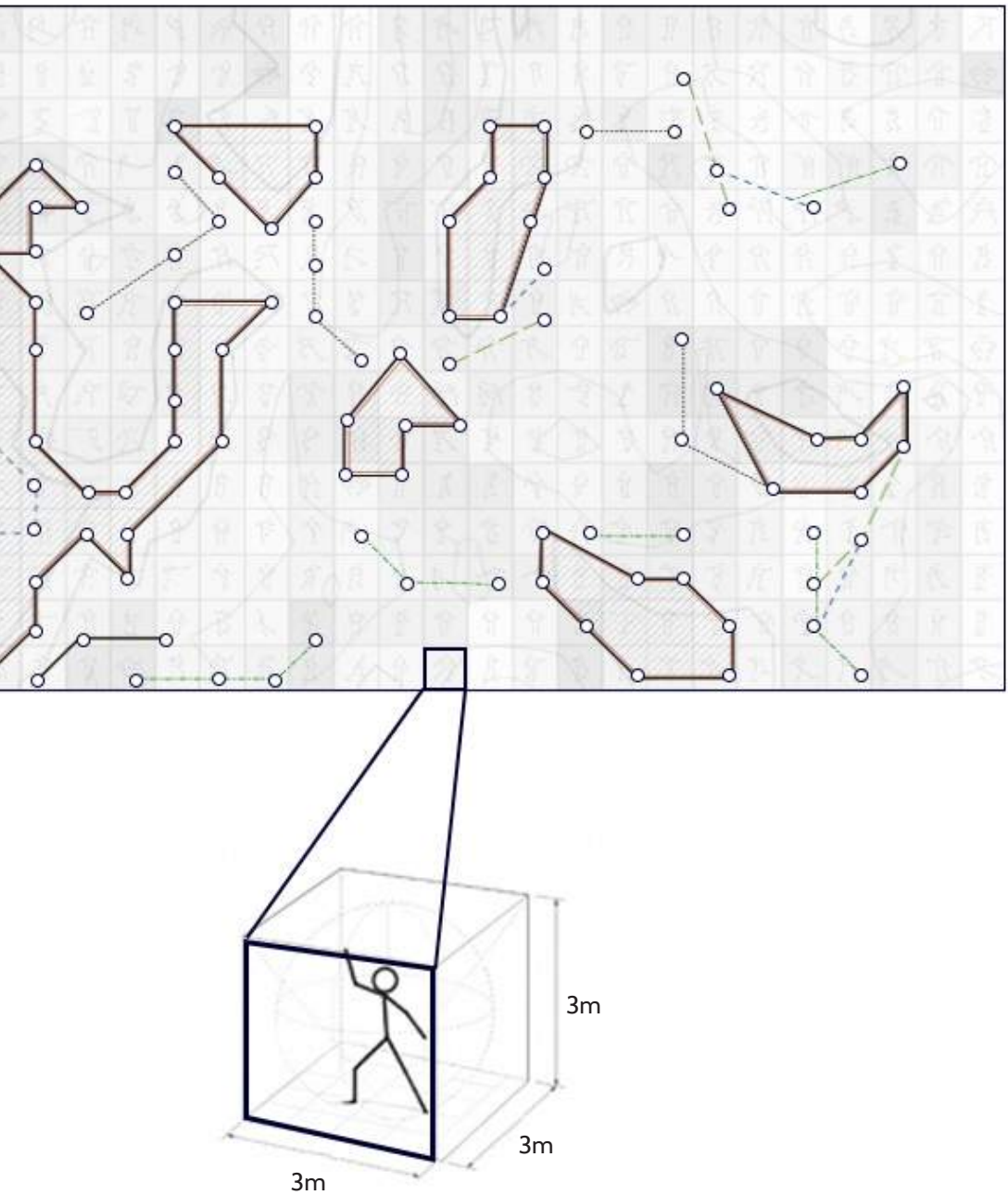

0.25
0.55
0.40
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0.25

PATH GENERATION

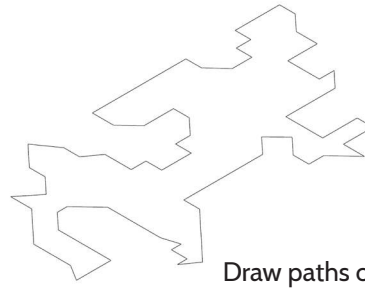


ground line
step line
wall line
handrail line
roof line
hand surface line
selected video

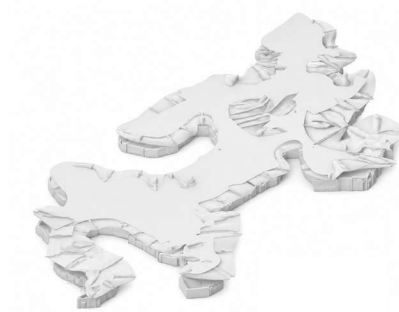
Paths are first drawn across the foot-movement heatmap. The energy intensity of each area then guides the drawing of paths on the maps assigned to different architectural elements. Each grid cell measures 3 × 3 metres, representing the maximum spatial range that a moving human body can occupy. The selected paths are translated into 3D models in Blender in real time, while the energy values of the music heatmap control the scale of the generated forms.


3m
3m
3m

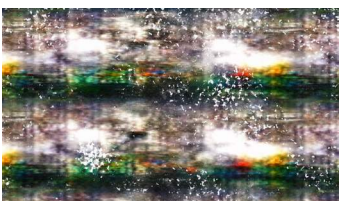
ELEMENT ASSEMBLY



Draw paths on element maps

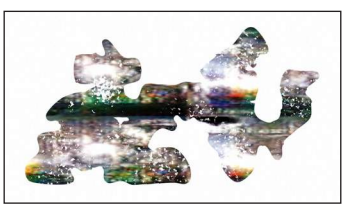


Combine generated elements





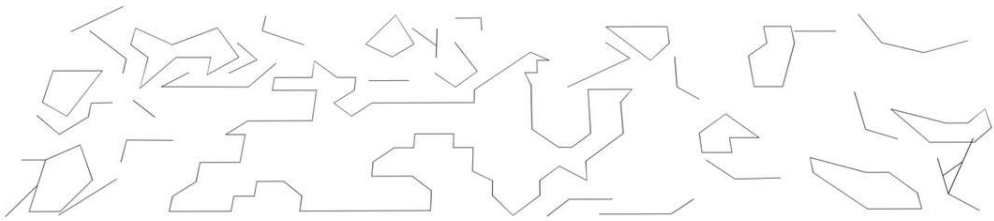
Refine the model





Apply materials

generation video link









FINAL ATMOSPHERE

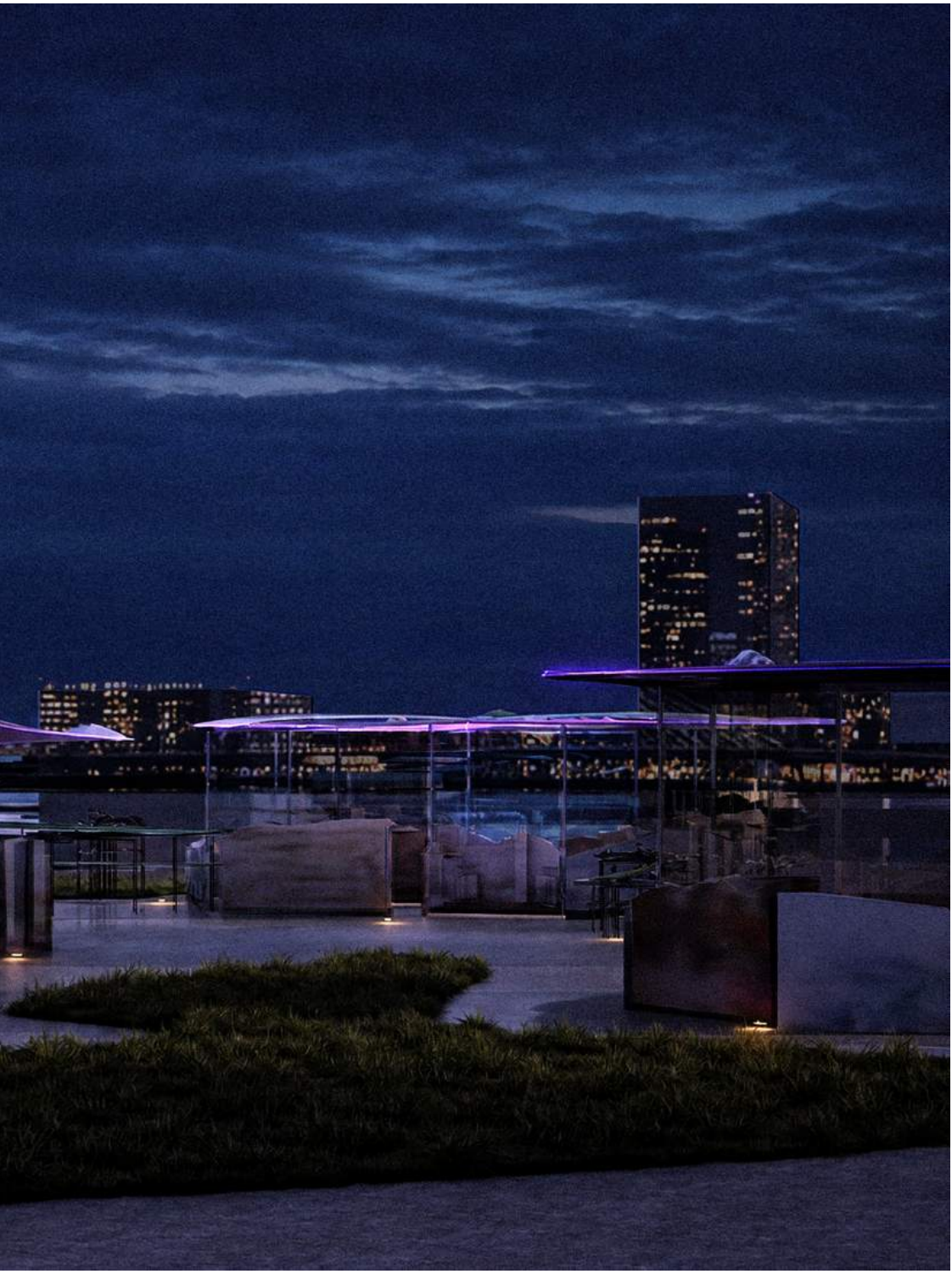


architectural walkthrough video



FINAL ATMOSPHERE





FINAL ATMOSPHERE







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School of
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